Problem 4

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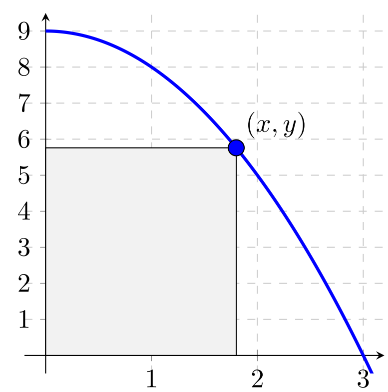
Problem 1 *A **rectangle in the first quadrant** is constructed by taking a point (x, y) on the graph of the function $f(x) = 9 - x^2$, drawing a line segment vertically downward to the x-axis and a line segment horizontally leftward to the y-axis, as in the picture below. Denote the **length of the base** (along the x-axis) of the rectangle by x, the **height of the rectangle** (along the y-axis) by y (in meters), and the **PERIMETER** of the rectangle by P. When $x = 2$ m (and only at that moment), the height y is shrinking at a rate of $\frac{1}{5}$ m/s. Find the value of $\left[\frac{dP}{dt}\right]_{x=2m}$ **by performing the steps below**.*



(a) ***Find a formula*** *for the height, y, of the rectangle as a function of x. (This is denoted as $y(x)$.)*

(b) ***Find the value*** *of $\left[\frac{dx}{dt}\right]_{x=2m}$.*

(c) ***Find a formula*** *for the perimeter, P, of the rectangle as a function of x. (This is denoted as $P(x)$.)*

(d) ***Find the value*** *of $\left[\frac{dP}{dt}\right]_{x=2m}$.*

(e) ***Write a sentence to explain*** *what the value found in (d) means about the rectangle. (Don't forget UNITS.)*